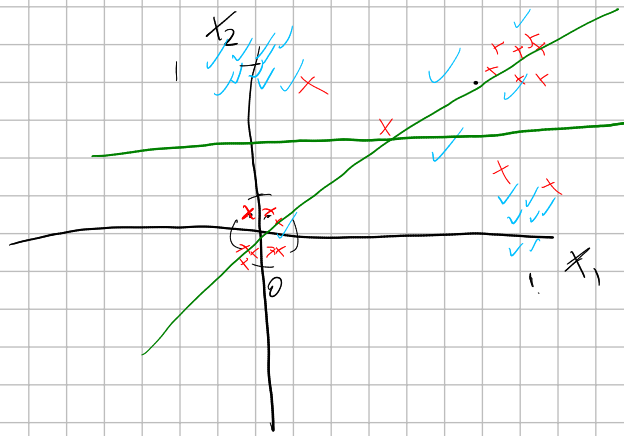




XOR

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	0

$$P(y=1) = \sigma(\beta_0 + \beta_1 x_1 + \beta_2 x_2)$$

$$\sigma(z) = e^z / (1 + e^z)$$

AI Winter (86:)

Radial Basis Functions

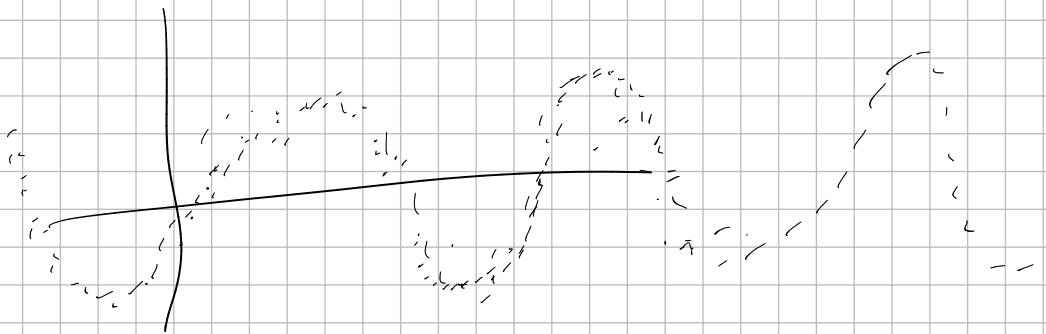
$$f(x) = \sum_{i=0}^m \beta_i f_i(x)$$

↳ RPoly

$$\sum_{i=0}^m \beta_i x^i$$

↳ Splines

$$\beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 (x - c_3)_+^3 + \beta_4 (x - c_4)_+^4$$



$x \in \mathbb{R}$

$$f_0(x) = 1$$

$$f_1(x) = x$$

$$f_2(x) = x^2$$

⋮

$$f_m(x) = x^m$$

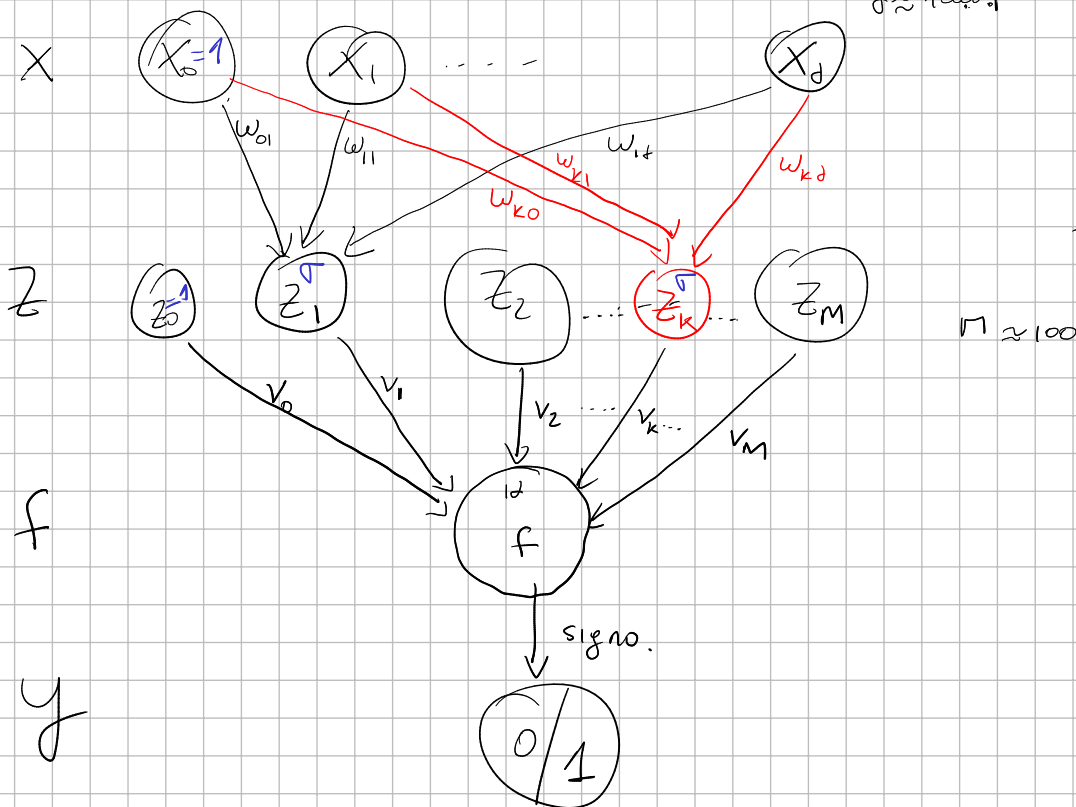
$$f(x) = \sum_{j=1}^m a_j \underline{f_j(x)}$$

↳ como elegirlos esor automáticamente?

1.) Métodos de kernel

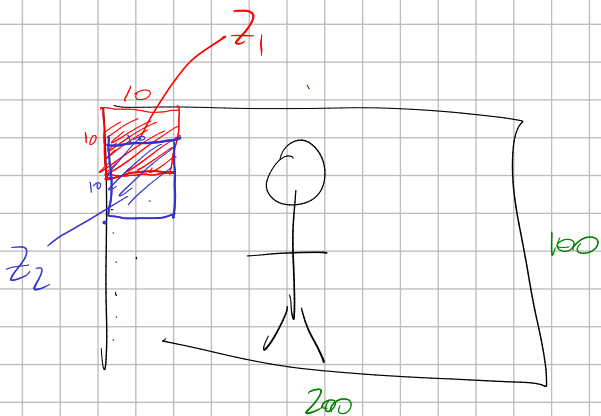
2.) Redes neurales (perceptron)

$$f(x) = \sum_{j=1}^m a_j \sigma(w_{j0} + w_j^T x)$$

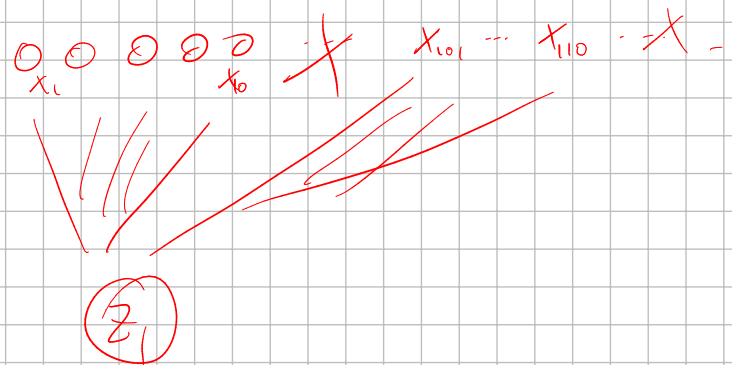


$$\sum_{k=1}^{M_1} b_k \sigma(\tilde{w}_{k0} + \tilde{w}_k^T z_k)$$

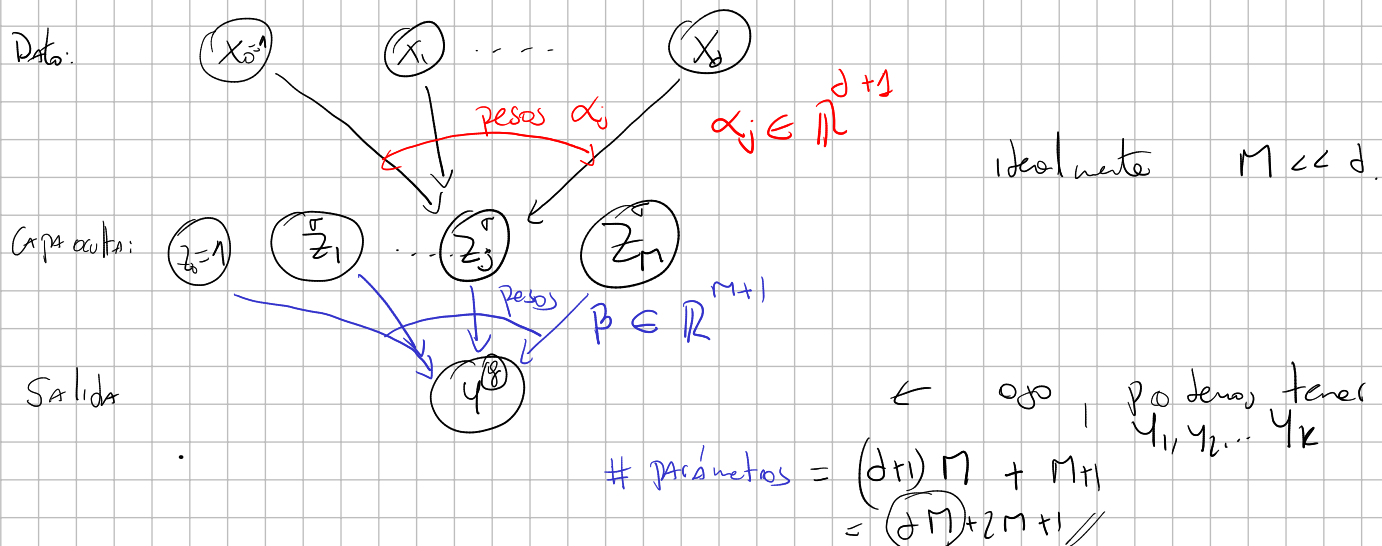
$$z_k = \sum_{j=1}^{M_0} a_j \sigma(w_{j0} + w_j^T x)$$



$$d = 40000$$



1) Formulaci3n matem3tica de un modelo con una capa oculta.



Matem3ticamente se puede escribir un modelo jer3rquico.

$$z_m = \sigma(\alpha_m^T x); \quad m=1, \dots, M \quad \sigma(u) = \frac{e^u}{1+e^u}; \quad u \in \mathbb{R}$$

$$y = g(\beta^T z); \quad g(u) \begin{cases} g(u) = u & \text{en regresi3n} \\ g(u) = \sigma(u) & \text{a clasificaci3n} \end{cases}$$

$$z = \begin{pmatrix} z_0 \\ z_1 \\ \vdots \\ z_M \end{pmatrix}$$

2) C3mo encontramos los pesos $\alpha_1, \dots, \alpha_M \in \mathbb{R}^{d+1}$ y $\beta \in \mathbb{R}^{M+1}$?

Sea $f_{\alpha, \beta}: \mathbb{R}^d \rightarrow \mathbb{R}$ la funci3n asociada a la red de par3metros α 's y β

- En regresi3n vamos a considerar p3rdida cuadr3tica $L(X_i, y_i) = (y_i - f_{\alpha, \beta}(x_i))^2$
- En clasificaci3n consideramos entropia

$$L(X_i, y_i) = - (y_i \log(\sigma(\beta^T z)) + (1-y_i) \log(1-\sigma(\beta^T z)))$$

$\uparrow$
 $\mathbb{R}^{d+1}$

ojo: $z = z(x_i) \leftarrow$ los valores en la capa oculta al evaluar en x_i

As3 minimizamos

$$\min_{\alpha, \beta} \frac{1}{n} \sum_{i=1}^n L(X_i, y_i)$$

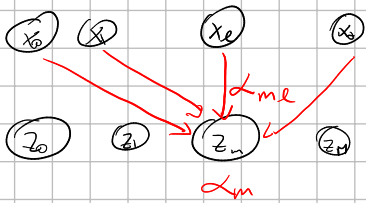
Calculamos el gradiente de $L(y_i, x_i)$ para un dato (x_i, y_i) fijo.
 en el caso de regresión (pérdida cuadrática)

$$L(y_i, x_i) = (y_i - f_{\alpha, \beta}(x_i))^2 \quad \text{ojo} \quad \begin{aligned} y_i &\in \mathbb{R} \\ x_i &\in \mathbb{R}^d \\ Z(x_i) &\in \mathbb{R}^{M+1} \\ \beta &\in \mathbb{R}^{M+1} \end{aligned}$$

$$= (y_i - \beta^T Z(x_i))^2$$

$$\frac{\partial}{\partial \beta_k} L(y_i, x_i) = -2 Z_k(x_i) (y_i - \beta^T Z(x_i)) \quad ; \quad k \in \{0, \dots, M\}$$

$$\frac{\partial}{\partial \alpha_{me}} L(y_i, x_i) = \frac{\partial}{\partial \alpha_{me}} \left(y_i - \beta_0 Z_0(x_i) - \beta_1 Z_1(x_i) - \dots - \beta_m Z_m(x_i) - \dots - \beta_M Z_M(x_i) \right)^2$$



↑
 A esta α_{me}

$$= 2(y_i - \beta^T Z(x_i)) \frac{\partial}{\partial \alpha_{me}} (y_i - \beta^T Z(x_i))$$

$$= -2(y_i - \beta^T Z(x_i)) \frac{\partial}{\partial \alpha_{me}} (\beta_m Z_m(x_i))$$

$$= -2(y_i - \beta^T Z(x_i)) \beta_m \frac{\partial}{\partial \alpha_{me}} Z_m(x_i)$$

$$= -2(y_i - \beta^T Z(x_i)) \beta_m \frac{\partial}{\partial \alpha_{me}} \sigma(\alpha_m^T x_i)$$

$\alpha_m = \begin{pmatrix} \alpha_{m0} \\ \alpha_{m1} \\ \vdots \\ \alpha_{me} \\ \vdots \\ \alpha_{mM} \end{pmatrix}$

APARTE

$$\alpha_m = \begin{pmatrix} \alpha_{m0} \\ \alpha_{m1} \\ \vdots \\ \alpha_{me} \\ \vdots \\ \alpha_{mM} \end{pmatrix} \quad x_i = \begin{pmatrix} x_{i0} \\ x_{i1} \\ x_{i2} \\ \vdots \\ x_{iM} \end{pmatrix}$$

$$= -2(y_i - \beta^T Z(x_i)) \beta_m (Z_m(x_i)) (1 - Z_m(x_i)) x_{iM}$$

(ver aparte)

$$\left(\alpha_m^T x_i = \alpha_{m0} x_{i0} + \alpha_{m1} x_{i1} + \alpha_{m2} x_{i2} + \dots + \alpha_{me} x_{iM} + \dots + \alpha_{mM} x_{iM} \right)$$

Resumen:

$$\frac{\partial}{\partial \beta_k} L(y_i, x_i) = -2 z_k(x_i) (y_i - \beta^T z(x_i))$$

$$\frac{\partial}{\partial \alpha_{me}} L(y_i, x_i) = -2 (y_i - \beta^T z(x_i)) \beta_m z_m(x_i) (1 - z_m(x_i)) x_{me}$$

$$k \in \{0, 1, \dots, M\}$$

$$m \in \{0, 1, \dots, M\}$$

$$l \in \{0, 1, \dots, d\}$$

3. Evidentemente hay que usar método del gradiente (gradient descent)

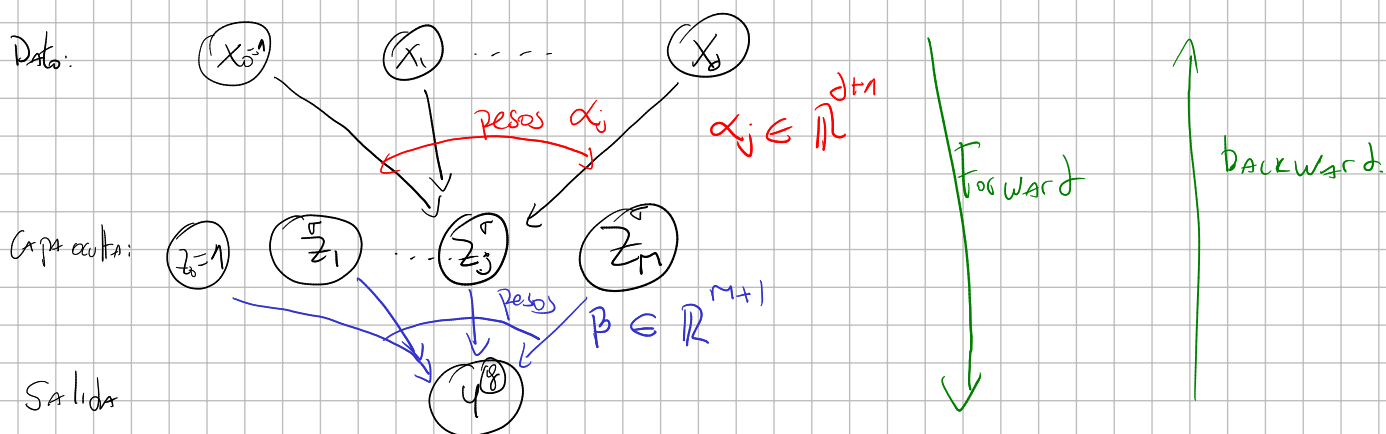
Elegimos $\alpha_1^{(0)}, \dots, \alpha_n^{(0)} \in \mathbb{R}^{d+1}$

$$\text{y } \beta^{(0)} \in \mathbb{R}^{M+1}$$

y así

$$\beta_k^{(r+1)} = \beta_k^{(r)} - \eta_r \frac{1}{n} \sum_{i=1}^n \underbrace{(-2 z_k(x_i) (y_i - \beta^{(r)T} z(x_i)))}_{\frac{\partial}{\partial \beta_k} L(x_i, y_i)} \bigg|_{\alpha^{(r)}, \beta^{(r)}} \quad k \in \{0, \dots, M\}$$

$$\alpha_{me}^{(r+1)} = \alpha_{me}^{(r)} - \eta_r \cdot \frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial \alpha_{me}} L(x_i, y_i) \bigg|_{\alpha^{(r)}, \beta^{(r)}}$$



En este contexto el algoritmo se llama backpropagation (BD)